

LegalMate.Pk



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Faculty of Computing

Riphah International University, Islamabad

Date: [date of final presentation]

Final Approval

This is to certify that we have read the report submitted by ***Muhammad Haroon Shahid (45669), Sameer Abid (47145), Sudais (46274)***, for the partial fulfillment of the requirements for the degree of the Bachelors of Science in Computer Science (BSSE). It is our judgment that this report is of sufficient standard to warrant its acceptance by Riphah International University, Islamabad for the degree of Bachelors of Science in Computer Science (BSSE).

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Declaration

We hereby declare that this document “**LegalMate.Pk**” neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers, especially our supervisors **Mr. Hassaan Siddique and Dr. Mansoor Alam**. If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

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Dedication

With sincere gratefulness to Allah Almighty for granting us the strength, patience, and willpower to complete this final year project. We would like to express our deepest appreciation to our respected supervisors for their valuable guidance, continuous support, and constructive feedback throughout the development of Legal Mate. Their mentorship played a vital role in shaping this project.

We are also thankful to our team members, whose commitment, collaboration, and hard work were essential in bringing this project to end. Their promise and partnership helped transform ideas into a real-world and meaningful solution.

Acknowledgement

First of all, we are obliged to Allah Almighty the Merciful, the Beneficent and the source of all Knowledge, for granting us the courage and knowledge to complete this Project.

We would like to thank our supervisors very much for always being there with their expert support, guidance, and positive criticism, besides, the supervisor's encouragement and mastery were crucial factors in our understanding and the enhancement of our work' s quality. We also owe a debt of gratitude to our teachers who were instrumental in laying the strong academic base upon which we built our final year project and for their continuous support. Friends and classmates are thanked for their collaboration, conversations, and moral support throughout the project, particularly during the idea development stage. Last but not least, we want to say that our families were always there for us, cheering up, being patient, and praying, thus giving us the strength, we needed to overcome this work successfully.

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Abstract

Text in 12-Point Size, Times New Roman, 1.5 Line Spacing.

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Abstract

An Abstract is a short summary of the work being reported. It should state: the purpose, findings, and conclusion of your work without commenting on or evaluating the work itself. **It should be only one paragraph at least half a page long.**

Chapter 1: Introduction

Due to the complexity of the language, access to legal information remains a significant challenge for the citizens of Pakistan. High consultation costs, and limited availability of legal professionals also plays a crucial role in this. Most people find it difficult to understand their rights, obligations and procedures without professional assistance. This problem becomes more severe for people from non-legal backgrounds, especially when dealing with court procedures and legal documentations etc.

To close this gap, **Legal Mate**, an AI-based legal aid system, is suggested. Through an intelligent conversational interface, the system seeks to give users automatic document generation, basic legal counsel, and simplified explanations of legal topics. Legal Mate aims to improve the general public's access to, comprehension of, and affordability of legal information by utilising massive language models, NLP techniques, and legal knowledge sources.

The goal of this project is to create an intelligent legal chatbot that can comprehend user enquiries written in common language, recognize what the user is asking, and offer relevant legal answers. The system supports many input methods, including text, voice, and optical character recognition (OCR), to make it more accessible and user-friendly.

1.1 Goals and Objectives

The primary **goal** of the Legal Mate project is to create an intelligent legal assistant that uses AI-driven technology to make it easier to obtain legal advice and information.

The main **objectives** of this project are listed below:

- To build an AI-based chatbot that can understand legal questions using Natural Language Processing (NLP).
- To explain legal terms, laws, and procedures in simple language so that people without a legal background can understand them easily.
- To create a conversational system that can give relevant and meaningful responses based on user questions.

- To add a feature that automatically generates basic legal documents such as applications, affidavits, and agreements.
- To allow users to interact with the system using different input methods, including text, voice, and images through OCR.
- To connect the chatbot with a legal knowledge base that includes laws, rules, and past legal cases.
- To test and evaluate the system based on how accurate, useful, and easy to use it is.
- To help users get basic legal guidance without always needing to consult a lawyer immediately.

1.2 Scope of the Project

Legal Mate's focuses to give users broad legal information and advice. Initially, the scope of this system is limited to handling legal queries related to **property disputes**, allowing the system to focus on a specific legal domain and provide more accurate and relevant guidance. Its goal is to make legal concepts, rights, and procedures easier for people to understand. The system does not represent users in court or provide legally binding rulings, nor is it intended to take the place of expert legal counsel.

The project includes:

- Development of an AI-based legal chatbot using a large language model trained on **property**-related data.
- Answering general legal questions related to property laws, rights, penalties, and legal procedures.
- Automatically generating basic legal documents for guidance purposes.
- Providing multilingual support where possible to make the system more accessible.
- Using NLP techniques to understand user intent and generate suitable responses.
- Designing an easy-to-use and interactive interface for users.

The project does not include:

- Representing users in court or handling litigation cases.
- Making real-time legal decisions or predicting court judgments.
- Replacing qualified lawyers or legal professionals.
- Processing highly sensitive or confidential legal information.

By clearly defining these boundaries, the project ensures that Legal Mate is used responsibly and ethically, while still being a reliable and helpful tool for users seeking basic legal information.

Chapter 2: Literature Review

2.1 Introduction

This chapter presents a review of existing research and related work relevant to the development of the Legal Mate system. The purpose of this literature review is to understand how similar problems have been addressed in the past, what methods and technologies have been used, and what limitations still exist in current solutions. Reviewing previous studies helps in identifying suitable approaches and avoiding duplication of work while building upon existing knowledge.

Recent advancements in artificial intelligence (AI) and natural language processing (NLP) have enabled the development of intelligent systems that can understand and respond to human language. These technologies have been widely applied in areas such as healthcare, education, and customer support. In the legal domain, researchers have explored AI-based systems to assist users by answering legal questions, analysing legal text, and generating documents. However, many of these systems are limited in scope, difficult for non-technical users to operate, or focused on general legal information rather than specific dispute types.

This chapter reviews research related to AI-based legal assistance systems, NLP techniques used in legal applications, large language models for question answering, automated legal document generation, and multimodal input methods such as voice and OCR. The aim is to analyse existing approaches, understand their strengths and weaknesses, and identify gaps that justify the need for the proposed Legal Mate system, particularly its focus on property-related legal disputes and user-friendly mobile deployment.

2.2 Background and Problem Elaboration

Legal systems are built on complex rules, formal language, and detailed documentation, which makes legal information difficult to understand for people without a legal background. Legal

documents often contain technical terms, long sentences, and references to laws and precedents that are not easily understandable by the general public. As a result, individuals facing legal issues frequently struggle to identify their rights, responsibilities, and the correct legal procedures.

Access to legal assistance is also limited by time, cost, and availability of legal professionals. Consulting a lawyer for basic legal guidance can be expensive and is not always immediately available, especially for individuals dealing with common disputes such as property-related matters. This creates a gap where users need quick and reliable legal information but are unable to obtain it easily through traditional legal services.

With the advancement of artificial intelligence (AI) and natural language processing (NLP), researchers have explored automated systems that can assist users by answering legal questions and processing legal text. Legal question answering systems aim to understand user queries written in natural language and retrieve or generate relevant legal responses. However, due to the complexity of legal language and the wide variety of legal domains, developing accurate and reliable legal assistance systems remains a challenging task.

Existing legal AI systems often suffer from several limitations. Many systems are rule-based, making them inflexible and difficult to scale to new legal domains. Others rely on large datasets that may not cover specific areas such as property disputes or local legal frameworks. In addition, many research systems focus on theoretical models and lack practical deployment in user-friendly platforms such as mobile applications. Limited support for multilingual input, voice-based interaction, and document understanding through OCR further reduces accessibility for a wide range of users.

These challenges highlight the need for an intelligent, accessible, and domain-focused legal assistance system. There is a clear requirement for a solution that can understand user queries in simple language, provide clear and relevant legal guidance, and support modern interaction methods. Addressing these issues forms the foundation for the development of the Legal Mate system, which aims to provide simplified legal assistance with an initial focus on property-related disputes.

2.3 Detailed Literature Review

2.3.1 Definitions

- **Artificial Intelligence (AI)**

Artificial Intelligence refers to computer systems designed to perform tasks that normally require human intelligence, such as reasoning and decision-making. In the legal field, AI is used to analyse legal text and assist users in understanding legal information.

- **Natural Language Processing (NLP)**

Natural Language Processing is a branch of AI that enables machines to understand and process human language. NLP techniques help legal systems analyse user queries and interpret complex legal language accurately.

- **Question Answering (QA) Systems**

Question Answering systems are designed to provide direct answers to questions asked in natural language. These systems use NLP and machine learning techniques to search information and return relevant responses.

- **Legal Question Answering (LQA)**

Legal Question Answering focuses on answering questions related to laws, regulations, and legal documents. LQA systems require high accuracy due to the sensitive and complex nature of legal information.

- **Deep Learning**

Deep learning is a machine learning approach that uses multi-layer neural networks to learn complex patterns from data. It is widely used in legal QA systems to improve understanding of legal language and context.

- **Large Language Models (LLMs)**

Large Language Models are advanced deep learning models trained on large text datasets to understand and generate human-like language. In legal applications, LLMs help produce more accurate and context-aware responses.

2.3.2 Related Research Work 1

Answering questions related to the legal domain is a complex task due to the intricate nature of legal language and the wide variety of legal document systems [1]. The authors explain that providing accurate answers to legal questions often requires specialized legal knowledge, which makes the task challenging even for human experts. To address this challenge, the study focuses on legal question answering (LQA) systems, which aim to generate answers to legal questions expressed in natural language using artificial intelligence and natural language processing techniques [1].

The study presents a comprehensive survey of existing legal question answering research rather than proposing a single implemented system. The authors review fourteen benchmark datasets used for legal QA and analyse state-of-the-art deep learning models applied in this domain [1]. According to the paper, legal QA systems typically use natural language processing to understand the meaning of a question and then apply machine learning and information retrieval techniques to locate the most relevant answers from large collections of legal documents. The survey highlights the growing use of deep learning and transformer-based architectures due to their ability to handle complex and unstructured legal text and to capture contextual information more effectively than traditional rule-based approaches [1].

The findings of the survey show that deep learning-based legal QA systems have achieved improved performance across different datasets and evaluation metrics [1]. The authors report that these models benefit from end-to-end learning and the ability to learn contextual relationships within legal text. However, the study also identifies several limitations, including

the need for large amounts of high-quality labelled data, domain-specific datasets, and challenges related to reliability, transparency, and trustworthiness of legal QA systems. Ethical concerns, data privacy issues, and the risk of incorrect or misleading answers are also highlighted as major challenges for real-world deployment [1].

In conclusion, this survey provides a detailed overview of existing legal question answering systems, datasets, and deep learning techniques, highlighting both their strengths and limitations [1]. While the reviewed approaches demonstrate the effectiveness of deep learning models in addressing legal QA challenges, the study indicates that many existing systems remain research-oriented and lack practical, user-friendly deployment. These limitations justify the need for applied systems such as Legal Mate, which aim to utilize legal QA techniques in a more accessible and domain-focused manner, particularly for property-related legal disputes.

2.3.3 Related Research Work 2

TBC.....

2.4 Literature Review Summary Table

The columns in the table depend upon your problem and should be specific to your project.

Table 1: History of Computing Devices
The summary of various computing devices invented in the past from 1833-1901 is presented here.

No.	Name, reference	Inventor	Year	Input	Output	Description
1.	Analytical Engine, [1]	Charles Babbage	1833	Punch cards	Printer, curve plotter, bell	First general purpose computer that had an arithmetical logic unit and could compute using conditional branching and loops. Also incorporated integrated memory.

2.5 Research Gap

2.6 Problem Statement

Chapter 3: Requirements and Design

Describe all modules of requirements and design in clear English text along with the necessary diagram and figures. Anyone reading your report should be able to reproduce your system/results after reading it.

For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions. Make sure no heading/subheading is blank. Write text to introduce each section as well.

Introduce sub-heading as:

3.1 Requirements

3.1.1 Functional Requirements

3.1.2 Non-Functional Requirements

3.1.3 Hardware and Software Requirements

3.2 Proposed Methodology

3.3 System Architecture

3.4 Use Cases

3.4.1 Sample Use Case Name Here

Name	Sample Use Case Name Here		
Actors	Admin, Business Owner, Store Manager		
Summary	The user shall provide their email and password on the login form and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records either added by any of the authorized users or added manually by a developer. The user must not already be logged in.		
Post-Conditions	The user’s session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
3	The user enters valid email and	4	The system verifies the email and

	password.		password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: <i>Incorrect email or password entered.</i>

3.5 Database Design (*Optional*)

3.6 Class Diagram (*Optional*)

3.7 Sequence diagram (*Optional*)

3.8 Any Other Artifact...

3.9 GUI Graphical User Interfaces (*Optional*)

This section should give the GUI dumps of each screen, with reference to the user. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers.

Chapter 4: Implementation and Test Cases

For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.

4.1 Implementation

Whatever implementation that you have done so far, please elaborate here.

Give clear details of the algorithms that were implemented along with the platform and the APIs which were used. **For FYP-1, this chapter can be changed to description of prototype developed.**

4.1.1 Implementation of First Component/Algorithm

Write implementation of first component of your system here.

4.2 Test case Design and description

This section will be added in FYP-II. Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail.

4.2.1 Sample Test case No.1

<Software component Name>			
<Reference>			
Test Case ID:	<i>Reference Number</i>	Test Date:	<i>Date</i>
Test case Version:	<i>Version number</i>	Use Case Reference(s):	<i>Relation to use cases</i>
Revision History:	<i>Refer to previous test case identity (if any)</i>		
Objective	<i>Need and scope of the testing</i>		
Product/Ver/Module:	<i>Refer to overall system being built and the place of this test case in it.</i>		
Environment:	<i>Necessary and desired properties of the test environment. (hardware/software)</i>		
Assumptions:	<i>Assumptions that might affect the testing process.</i>		
Pre-Requisite:	<i>Necessary condition that needs to be fulfilled prior to the test case.</i>		
Step No.	Execution description	Procedure result	
	<i>Events being tested.</i>	<i>Mention software response.</i>	
Comments:			
☐ Passed ☐ Failed ☐ Not Executed			

4.2.2 Sample Test case No.2

.

.

4.3 Test Metrics

Summarize here the common ground of attributes of test case metrics.

4.3.1 Sample Test case Metric.No.1

Metric:	Purpose
Number of Test Cases:	Total number of test cases that you have developed for your system.
Number of Test Cases Passed:	The number of test cases that successfully passed
Number of Test Cases Failed:	The number of test cases that failed
Test Case Defect Density:	(No of test cases failed * 100) No of test cases executed
Test Case Effectiveness:	No of defects detected using test cases *100 Total number of defects detected
Traceability Matrix:	Traceability is the ability to determine that each feature has a source in requirements and each requirement has a corresponding implemented feature.

4.3.2 Sample Test case Metric.No.2

4.3.3 Sample Test case Metric.No.3

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Chapter 5: Experimental Results and Analysis

This chapter will be added in FYP-II. Give proper analysis and discussion of experimental results (in plain English text) along with tables of results. **For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.**

Chapter 6: Conclusion and Future Directions

This chapter is mandatory. Give conclusions and summary of the work done. What were your findings and what were the results? Discuss in detail whether the scope of your project was entirely covered or not and whether the objectives of the project were met or not. What challenges did you face and what has been left out and why?

Sum up all the conclusions of all the chapters here to make a conclusion chapter. Do not repeat any text, just summarize it in different words.

Give recommendations for future work also. How your project can be further enhanced or improved? Future recommendations if someone wants to work on it. **For FYP-1 it is mandatory to list down a plan of the work to be done for FYP-2.**

References

List all important sources of information which have been consulted for this project

Appendix

Appendix A: Guidelines

This section should include all supporting information from the project that was not included in the body of the report. You should include surveys, complex statistical calculations, certain detailed tables and other such information in an appendix. The information presented in this section is important to support the work presented in the body of the report but would make it more difficult to read and understand if presented within the body of the report.

Cite the appendix items in the report narrative (write "see Appendix A") and organize appendices (e.g., Appendix A, Appendix B,

Any tables, figures, forms, or other materials that are not totally central to the analysis but that need to be included are placed in the Appendix.

Appendix B: Heading of Sample Appendix B

Following is a sample code with “code” style format.

```
Void SampleFunction(){  
    Print "Hello World.";  
}
```

Formatting Guidelines

This document also serves as style guide for final year project reports. In order to give a similar high-quality appearance to all final year software project reports this template uses a collection of predefined Microsoft Word formatting styles. **These styles should be used without modification or replacement.** Font in the document is “*Time New Roman*”. This template provides following styles:

- **Title** – the main title style
- **Title2** – the subtitle style
- **Body Text** – style for paragraphs
- **Caption** – the style for a figure or table caption
- **Table Description** – the style for description of table, it must be added after caption.
- **Figure Description** - the style for description of figure, it must be added after caption.
- **Code** – the style for program source code

```
int x = 10; // Writing important code
```
- **Table Header Row** – Style for the header row of table
- **Table Grid** – the style for the data rows in the tables
- **Reference** – The style for references
- **Bullets** – The style for the bullet lists
- **Numbered List**– Style for numbered lists

All Heading styles with different level numbers are listed below.

Chapter 1: Heading 1

1.1 Heading 2

1.1.1 Heading 3

1.1.1.1 Heading 4

1.1.1.1.1 Heading 5

1.1.1.1.1.1 Heading 6

1.1.1.1.1.1.1 Heading 7

1.1.1.1.1.1.1.1 Heading 8

1.1.1.1.1.1.1.1.1 Heading 9

Tables and Figures

Tables and figures should be centered horizontally. The caption button should be used to insert caption for both the figures and tables. All figures and tables must be numbered properly. Always refer to tables and figures according to their numbers. A table or figure can be cited as follows: ‘see Table1’ or ‘as shown in Table1’. The caption of table should be centered above the table and figure caption should be centered below the figure. Place the tables/figures close to their reference. Use “Table Header Row” and ‘Table Grid’ style for table’s header and data rows respectively. It is compulsory to provide brief description of table/figure after its caption. Styles for table and figure descriptions are “Table Description” and “Figure Description” respectively.

Press Ctrl+Shift+S to see list of styles mentioned above. Figure 1 shows the Apply Style window displaying the list of styles. Select any text then press Ctrl+Shift+S, the Apply Style window will show you the current style applied on that text and if required, you can change the style by selecting any other style from the “Style Name” dropdown.

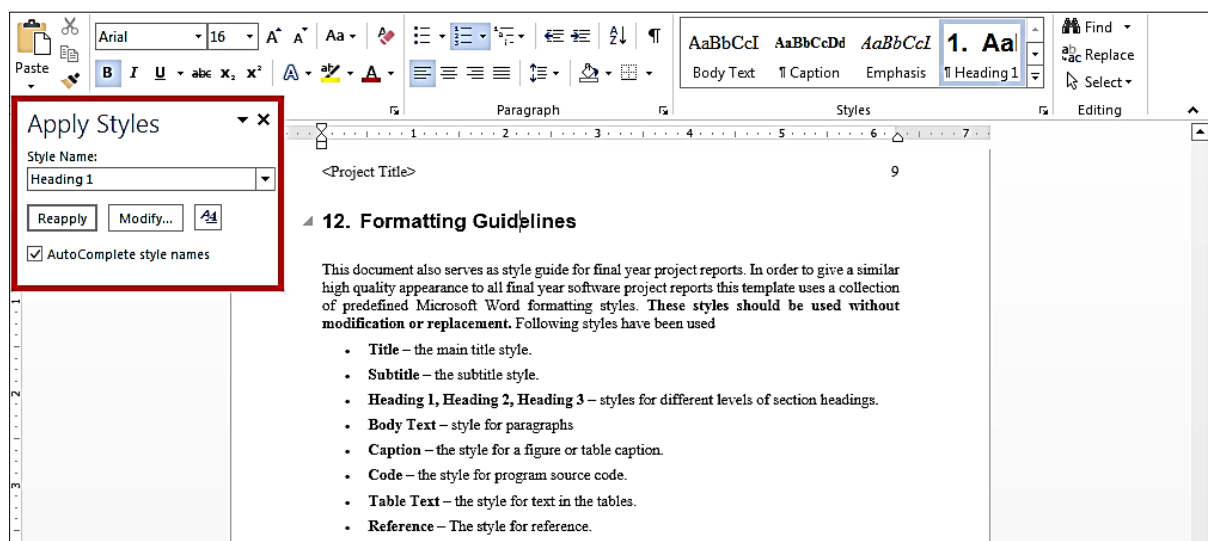


Figure 1: List of Styles

This is brief description of above figure.

Table 2: This is Sample table caption

This is brief description of following Table.

Header row	Header row	Header row	Header row
Row1 col1	Row1 col2	Row1 col3	Row1 col4
Row2 col1	Row2 col2	Row2 col3	Row2 col4

Table 3: This is Sample table caption

This is brief description of following Table.

Header row	Header row	Header row	Header row
Row1 col1	Row1 col2	Row1 col3	Row1 col4
Row2 col1	Row2 col2	Row2 col3	Row2 col4

Equations

Use equation editor to write equations in this report. Use last button of the custom tool bar to invoke equation editor. Similar to tables and figures, equations should also be aligned centered horizontally. Number all equations and insert them in parenthesis. Below is a sample equation and its reference number. An equation can be referenced like this: ‘it is clear from (1)’.

$$\sum_{\forall v \in V(G)} \deg(v) = 2|E(G)| \quad (1)$$

Header/Footer

Notice the headers in this document, before Introduction (i.e. the main content of this document) page numbers are in roman numerals. The page numbers of the actual content start with Arabic numerals i.e. 1, 2, 3 and so on. All of the **odd numbered pages** contain title of your project while the **even numbered pages** contain the section heading (i.e. chapter’s name) in the headers.

Other Formatting Guidelines

- Keep 2-4 GUIs in one page. Consume as much space as possible. Do not leave most of page blank unnecessarily.
- Do not break tables (or use cases) in multiple pages unless the table is too large to fit in one page.
- Re-arrange the content i.e., text, images, and tables properly to meet above two guidelines.

References

Always refer to the source of information by inserting the reference number in square brackets like this [5]. The reference numbers can either be added at the end of the sentence or within the sentence without changing the punctuation of sentence. A reference can also be cited as follows: ‘as Ruskey [2] mentioned’. List each source only once on your reference page.

- | |
|--|
| <p>[1] B. Klaus and P. Horn, <i>Robot Vision</i>. Cambridge, MA: MIT Press, 1986.</p> <p>[2] L. Stein, “Random patterns,” in <i>Computers and You</i>, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.</p> <p>[3] R. L. Myer, “Parametric oscillators and nonlinear materials,” in <i>Nonlinear Optics</i>, vol. 4, P. G. Harper and B. S. Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.</p> <p>[4] M. Abramowitz and I. A. Stegun, Eds., <i>Handbook of Mathematical Functions</i> (Applied Mathematics Series 55). Washington, DC: NBS, 1964, pp. 32-33.</p> <p>[5] E. F. Moore, “Gedanken-experiments on sequential machines,” in <i>Automata Studies</i> (Ann. of Mathematical Studies, no. 1), C. E. Shannon and J. McCarthy, Eds. Princeton, NJ: Princeton Univ. Press, 1965, pp. 129-153.</p> <p>[6] Westinghouse Electric Corporation (Staff of Technology and Science, Aerospace Div.), <i>Integrated Electronic Systems</i>. Englewood Cliffs, NJ: Prentice-Hall, 1970.</p> <p>[7] M. Gorkii, “Optimal design,” <i>Dokl. Akad. Nauk SSSR</i>, vol. 12, pp. 111-122, 1961 (Transl.: in L. Pontryagin, Ed., <i>The Mathematical Theory of Optimal Processes</i>. New York: Interscience, 1962, ch. 2, sec. 3, pp. 127-135).</p> <p>[8] G. O. Young, “Synthetic structure of industrial plastics,” in <i>Plastics</i>, vol. 3, <i>Polymers of Hexadromicon</i>, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.</p> |
|--|

Figure 2: IEEE Reference style

This figure represents the styling information for adding references in IEEE format

Following is a list of sample reference for various typed of sources in IEEE format.

- [1] A. Abdallah, B. Piryani, and A. Jatowt, "Exploring the state of the art in legal QA systems," *Journal of Big Data*, vol. 10, no. 127, 2023, doi: 10.1186/s40537-023-00802-8.
- [2]
- [3] S.K. Kenue and J.F. Greenleaf, "Limited angle multifrequency diffraction tomography," *IEEE Trans. Sonics Ultrason.*, vol. SU-29, no. 6, pp. 213-217, July 1982. **//Format for Journal Article**
- [4] B. Tsikos, "Segmentation of 3-D scenes using multi-modal interaction between machine vision and programmable mechanical scene manipulation," Ph.D. dissertation, Univ. of Pennsylvania, BCE Dept., Philadelphia, 1987. [Add if applicable: University Microfilms, Inc., University of Michigan, Ann Arbor, Michigan.] **//Format for Dissertation or thesis**
- [5] R. Finkel, R. Taylor, R. Bolles, R. Paul, and J. Feldman, "An overview of AL, programming system for automation," in *Proc. Fourth Int. Joint Conf Artif. Intell.*, pp. 758-765, Sept. 3-7, 1975. **//Format for Proceedings paper**
- [6] "Technology threatens to shatter the world of college textbooks, *The Wall Street Journal*, vol 91, pp. A1, A8, June 1, 1993. **//Format for Newspaper article**
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